

HerGuardian AI

AI-Powered Preventive Healthcare Companion for Women

TECHNICAL REPORT

Live Deployment: her-guardian-ai-app-updated.vercel.app

Report Date: 15 July 2026

1. Project Overview

HerGuardian AI is an AI-powered preventive healthcare companion built for women, bringing health assessments, mental wellness support, doctor booking, a dynamic health calendar, testing-agency access, emergency alerts, educational resources, payments, and an AI chatbot into a single web platform.

The platform focuses on screening support, health education, preventive guidance, and timely awareness rather than diagnosis. It helps users understand health patterns, generate reports, and take safer next steps such as booking appointments or consulting healthcare professionals.

1.1 Core Modules

- AI health dashboard — health score, mental wellness, activity status, reports, recommendations, and emergency access
- Health Assessment — personalized risk score, condition-wise analysis, recommendations, downloadable PDF report
- Mental Wellness — mood tracking, PHQ-9 and GAD-7 screening, notes, reports
- Doctor Booking — gynecologists, general physicians, endocrinologists, cardiologists, breast specialists, bone specialists, mental health counsellors
- Weekly Checkup — regular health check-ins and trend analysis
- Health Calendar — dynamic tracking for periods, appointments, lab tests, and reminders
- Testing Agencies — Mumbai-focused blood tests, women's health panels, PCOS screening, body checkups, home sample collection
- Facts & Wellness Care — educational content, Burnout & Stress, Breast Health, and Bone Health modules
- AI Chatbot — educational guidance on symptoms, periods, PCOS, tests, appointments, stress, and preventive care
- Emergency Alerts — SMS, call, and email-ready flows
- Support & Payments — Razorpay-ready checkout (UPI, cards, net banking, QR) with partner discount coupons

2. Technology Stack

Layer	Technologies
Frontend	Next.js (App Router), React, TypeScript
Styling	Tailwind CSS, shadcn-style UI components, Lucide icons
Animation	Framer Motion
Charts	Recharts
Forms & Validation	React Hook Form, Zod
Database	PostgreSQL (Neon)
ORM	Drizzle ORM
Authentication	Better Auth / custom auth routes, Google OAuth
AI Chatbot	OpenAI / Gemini / Puter.js-compatible chat API
SMS, OTP & Calling	Twilio
Payments	Razorpay Checkout
Reports	jsPDF, html2canvas
Deployment	Vercel
Version Control	GitHub

3. Model, Runtime & AI Integration

The application uses a custom-trained machine learning model for health score and mental wellness score prediction, trained on the datasets described in the Datasets section (PHQ-9, NHANES-derived Women's Health Indicators, PCOS clinical data, and BRFSS lifestyle data). This model achieves 93% accuracy and powers the platform's health and wellness scoring logic.

Separately, the chatbot and AI-guidance layer are connected to an external, cloud-hosted large language model provider.

3.1 Supported AI Providers

- OpenAI-compatible API
- Gemini API
- Puter.js-compatible AI chat API

3.2 Runtime Environments

- Next.js server runtime (API routes, server components)
- Browser runtime for client-side interaction and rendering
- External cloud AI runtime for chatbot inference
- Local inference runtime for the custom health/wellness scoring model (runs within the application server)

3.3 Optimization Techniques

No custom quantization is applied, since no local model ships with the project. Application-level optimizations include:

- Next.js production build optimization and static rendering where possible
- Component-based rendering to minimize re-renders
- API route separation for clear service boundaries
- Lightweight chart rendering via Recharts
- External API calls made only when required (on-demand chatbot invocation)
- Browser-based PDF generation triggered only on explicit user request

3.4 Custom Wellness and Health Scoring Model

- Purpose: predicts health score and mental wellness score from user assessment, mood, and symptom inputs
- Training data: PHQ-9 scoring framework, NHANES-derived Women's Health Indicators dataset, PCOS clinical dataset, BRFSS lifestyle dataset
- Accuracy: 93%

3.5 Model Size

- Custom health/wellness scoring model: lightweight tabular model — typically a few hundred KB to low single-digit MB, consistent with a classical ML model trained on datasets of this scale (hundreds to a few thousand rows)
- The chatbot relies entirely on external hosted model APIs, so no additional local model size applies

4. Performance & Resource Metrics

4.1 Estimated Latency

Operation	Estimated Latency
Local form-based assessment	< 1 second
Calendar and local UI updates	< 1 second
PDF report generation	1 – 3 seconds (browser & report size dependent)
Chatbot response	1 – 5 seconds (provider & network dependent)
Razorpay order creation	1 – 3 seconds

4.2 Compute Usage

- CPU — browser rendering, form validation, scoring logic, PDF generation, JavaScript execution
- GPU — browser rendering and animation acceleration only
- NPU — not used
- Cloud AI inference — fully offloaded to the external AI provider

4.3 Peak Memory Usage (Browser)

Scenario	Approx. Memory
Normal dashboard usage	150 – 300 MB
Charts and wellness pages	250 – 500 MB
PDF generation	300 – 700 MB (report-size dependent)

4.4 Tested Device Specification

Parameter	Value
Operating System	Windows
Browser	Chrome / Edge
Runtime	Node.js
Deployment	Vercel
Database	Neon PostgreSQL
Network	Wi-Fi / broadband
Recommended RAM	4 GB or higher
Internet	Required for all cloud-backed services

5. Feature Walkthrough — Sample Inputs & Expected Outputs

This section documents each core module with its sample input, expected behaviour, and the actual application output captured from the live deployment.

5.1 Landing Page & Dashboard Entry

The landing page introduces the platform's value proposition — early risk detection, personalized guidance, and continuous wellbeing support — and routes the user into the assessment flow.

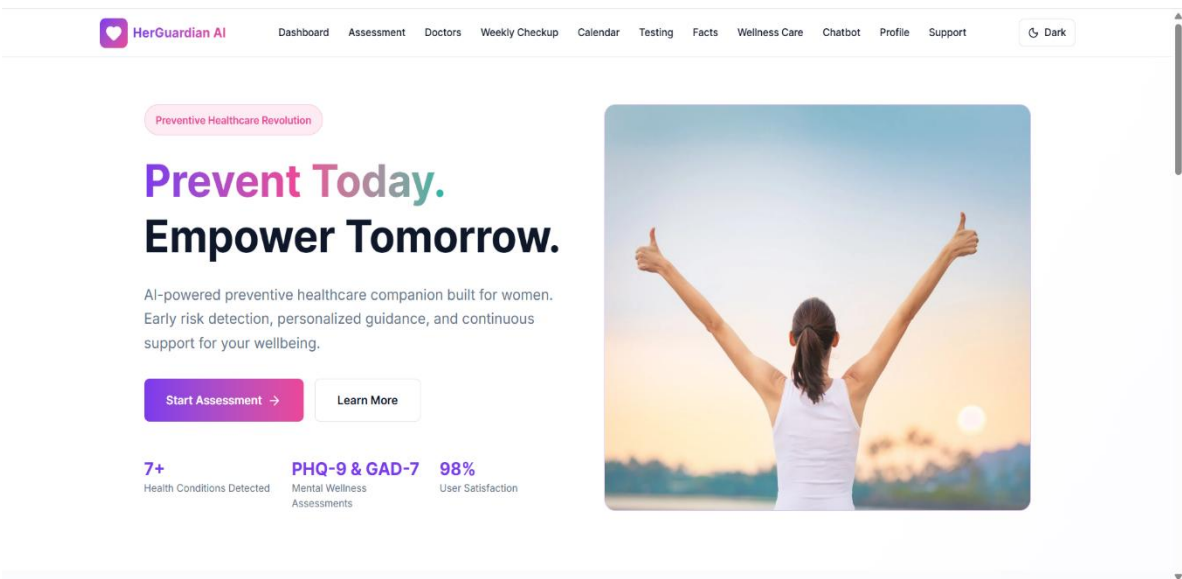


Figure 1 — HerGuardian AI landing page with entry point into the assessment flow.

5.2 Health Assessment

Sample Input: Sleep: 6 hrs · Stress: 7/10 · Mood: Low · Activity: Moderate · Symptoms: Fatigue, anxiety, irregular periods

Expected Output: A generated health score, condition-wise risk levels, personalized recommendations, and a downloadable PDF report, with professional consultation suggested for moderate/serious concern.

Actual output — the assessment engine produced a health score of 67 ("Needs Monitoring"), flagged PCOS as High Risk and Thyroid as Moderate Risk, and surfaced two targeted recommendations plus a safety disclaimer to confirm findings with a qualified doctor:

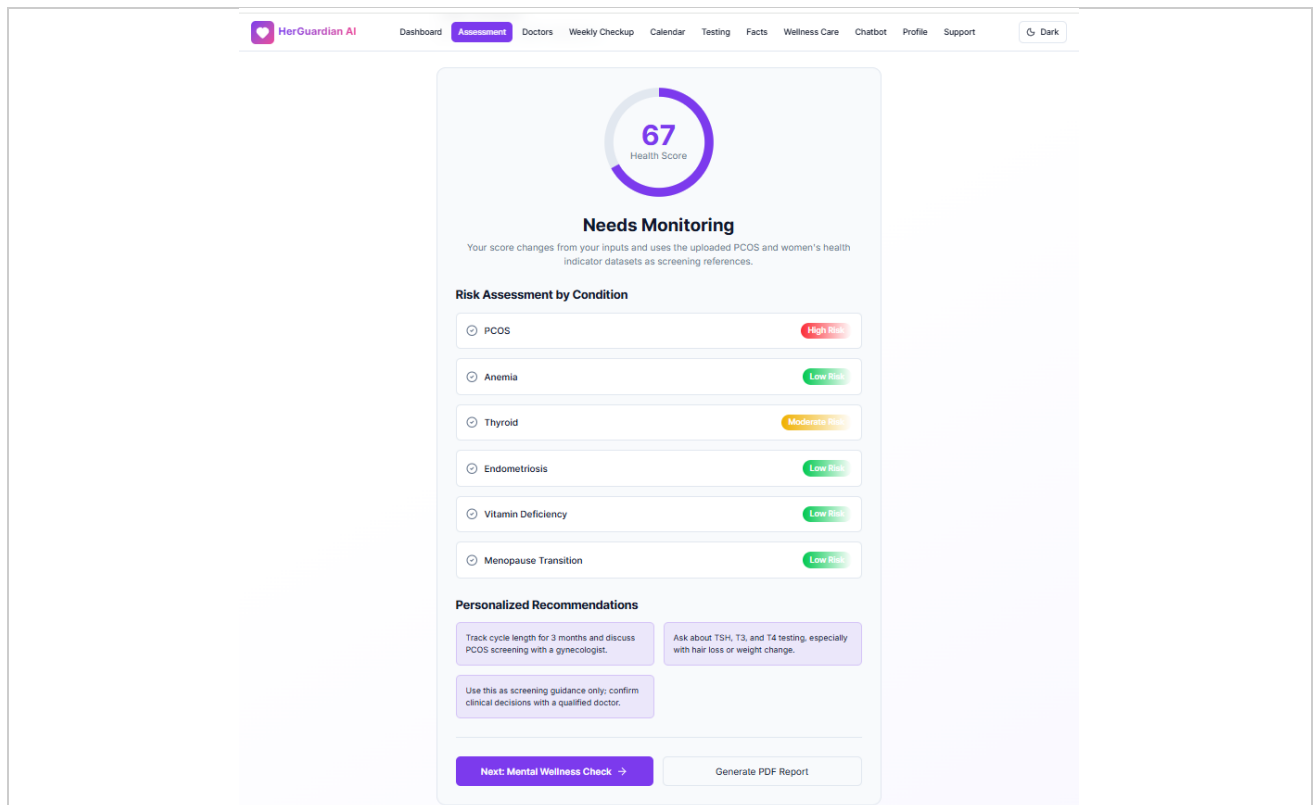


Figure 2 — Health Assessment result: score, risk-by-condition breakdown, and recommendations.

5.3 Doctor Booking

Sample Input: Specialty: Gynecologist · Date: 20 July 2026 · Reminder: Enabled

Expected Output: Booking flow opens, selected doctor details are shown, reminder information is prepared, and the calendar can reflect the appointment.

The Doctors module lists verified providers with specialty, experience, rating, distance, and consultation fee, with location-based sorting and a Book Now / Call action per listing:

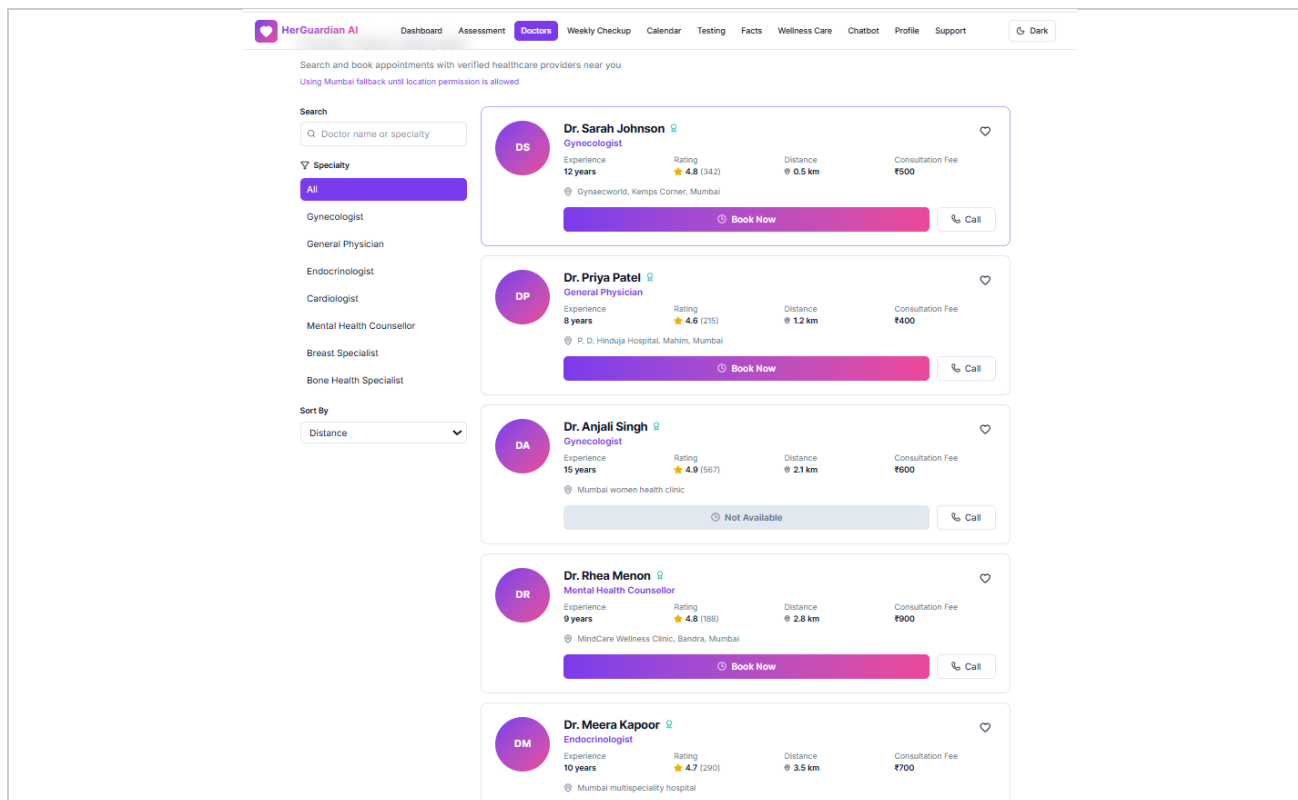


Figure 3 — Doctor directory filtered by specialty, sorted by distance (Mumbai fallback location).

Once a slot is confirmed, a toast notification confirms the booking and enables reminder/call alerts on the registered mobile number:

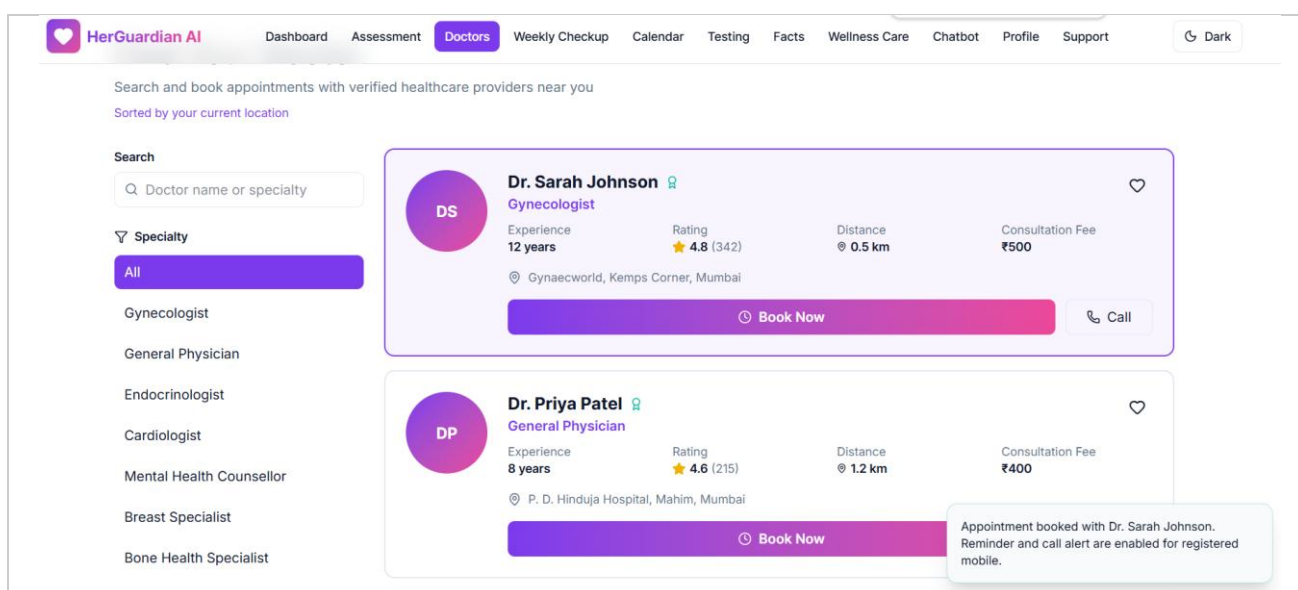


Figure 4 — Appointment confirmation toast after booking Dr. Sarah Johnson (Gynecologist).

5.4 Testing Agency Booking

Sample Input: Test: PCOS profile · Mode: At centre · Date: 17-07-2026 · Coupon: FIRSTCHECK50 · Payment: Pay at centre

Expected Output: Testing request details are prepared, discount information is shown, and the payment flow can continue through Razorpay.

The booking modal captures test type, date, time, payment mode, coupon code, and testing mode, confirming the applied discount before submission:

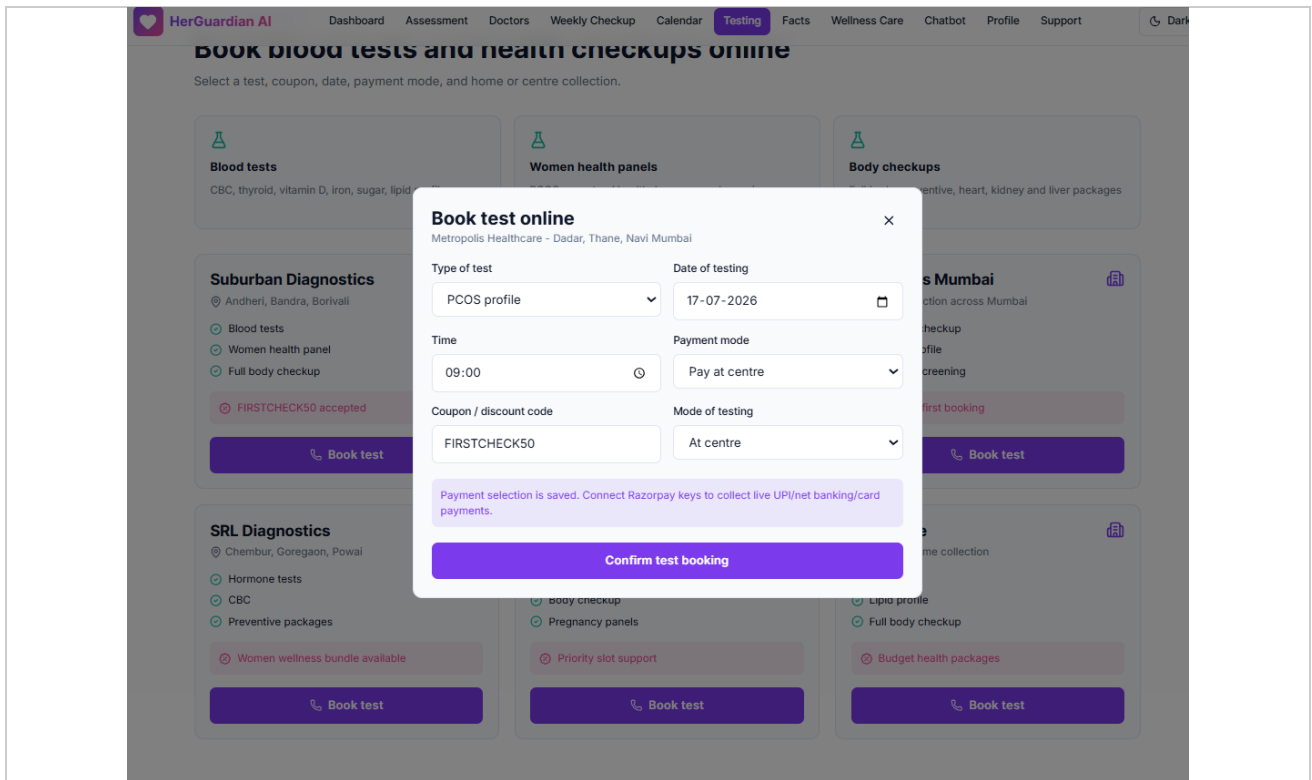


Figure 5 — Test booking modal (Metropolis Healthcare) with FIRSTCHECK50 coupon applied.

5.5 Health Calendar

Sample Input: Cycle length: 28 days · Period length: 5 days · Appointment: 20 July 2026 · Lab test: 22 July 2026

Expected Output: The calendar updates dynamically; period, appointment, and lab-test events appear in the 28-day view, alongside an upcoming-tasks summary.

Cycle settings are reflected live across a rolling 28-day grid, merging period-tracking days with the doctor appointment and lab test booked in earlier steps:

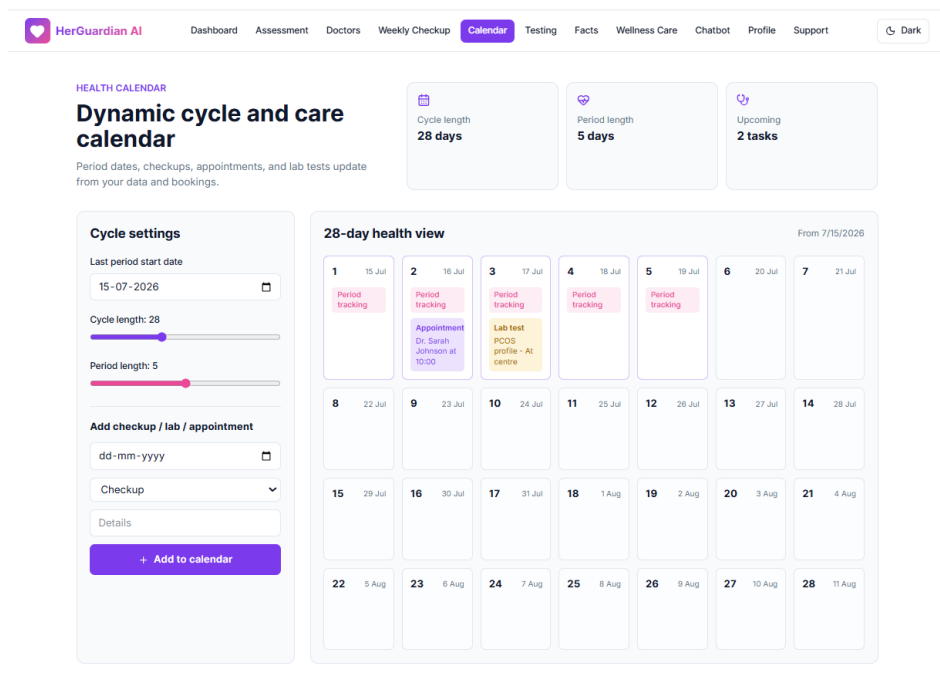


Figure 6 — Dynamic health calendar showing period tracking, appointment, and lab-test entries.

5.6 Wellness & Preventive Care

Three guided modules provide structured screening support without making diagnostic claims: Burnout & Stress, Breast Health, and Bone Health.

Burnout & Stress — the assessment combines context, sleep, workload, emotional exhaustion, caregiving burden, and lifestyle sliders into a wellness estimate:

The screenshot displays the HerGuardian AI interface for the Burnout & Stress module. The top navigation bar includes links to Dashboard, Assessment, Doctors, Weekly Checkup, Calendar, Testing, Facts, Wellness Care (active), Chatbot, Profile, and Support. A Dark mode toggle is also present.

GUIDED MODULE: Burnout & Stress
Workload, caregiving, sleep, recovery, and emotional exhaustion screening support.

Day Status:

- Calm day:** Good sleep, lighter load, more recovery.
- Busy but managing:** Some stress, reduced breaks, okay functioning.
- Overloaded:** High load, poor recovery, needs attention.

A. Burnout assessment

Progress: 1 Context, 2 Sleep, 3 Workload, 4 Emotions, 5 Lifestyle

Inputs:

- Age range: 25-34
- Employment status: Employed
- Work arrangement: Hybrid
- Caregiving responsibilities: 3/10
- Household workload level: 5/10

Current estimate: 38 (Wellness estimate: Mild)

This is a wellness burnout-risk estimate based on sleep, workload, emotional exhaustion, recovery, caregiving burden, and lifestyle inputs. It is not a diagnosis.

Factor	Score
Sleep	24
Workload	32
Emotional Exhaustion	42
Recovery	58
Caregiving Burden	26
Lifestyle	52

Completeness: 100%

B. Daily check-in

Date: 15-07-2026

Inputs:

- Mood: 5/10
- Energy: 5/10
- Stress: 5/10
- Sleep duration: 7
- Sleep quality: 5/10

D. Trends

No daily check-ins saved yet.

Figure 7 — Burnout & Stress module: wellness estimate of 38 ("Mild") with factor breakdown.

Breast Health — a self-awareness guide, personal reminder plan, and structured symptom log support early, doctor-ready preparation:

Figure 8 — Breast Health module: reminder plan, self-awareness guide, and symptom log.

Bone Health — age, height, weight, menopause status, and lifestyle factors feed a bone-health risk estimate with protective/risk factor callouts:

Figure 9 — Bone Health module: risk estimate of 6/100 ("Low") with key and protective factors.

5.7 AI Chatbot

Sample Input: "I have period cramps. What can I do?" (tested here with "what is PCOS?")

Expected Output: Educational guidance and safe self-care suggestions are displayed, with professional help recommended for severe or unusual symptoms.

On open, the chatbot introduces itself as a non-diagnostic educational assistant and lists urgent red-flag symptoms that warrant immediate care:

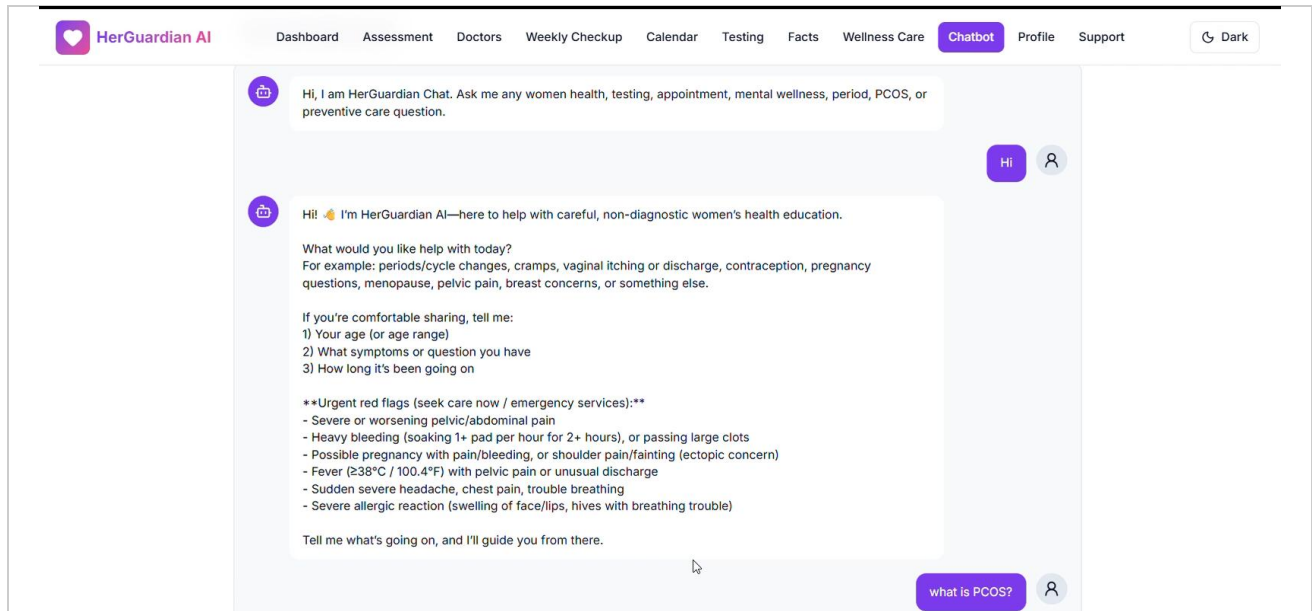


Figure 10 — Chatbot introduction with scope, guiding questions, and urgent red-flag list.

For the query "what is PCOS?", the chatbot returns a structured, plain-language explanation covering symptoms, underlying mechanism, and diagnostic approach:

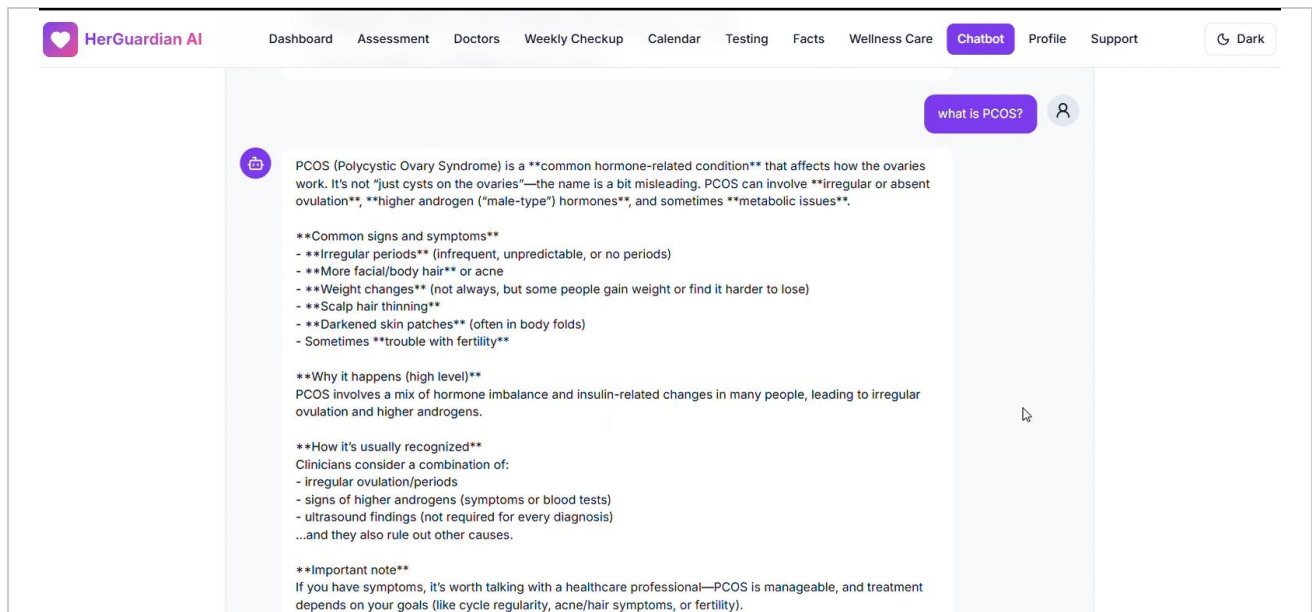


Figure 11 — Chatbot response explaining PCOS: symptoms, causes, and how it is recognized clinically.

The response concludes with an explicit recommendation to consult a healthcare professional and a condition-specific list of urgent red flags:

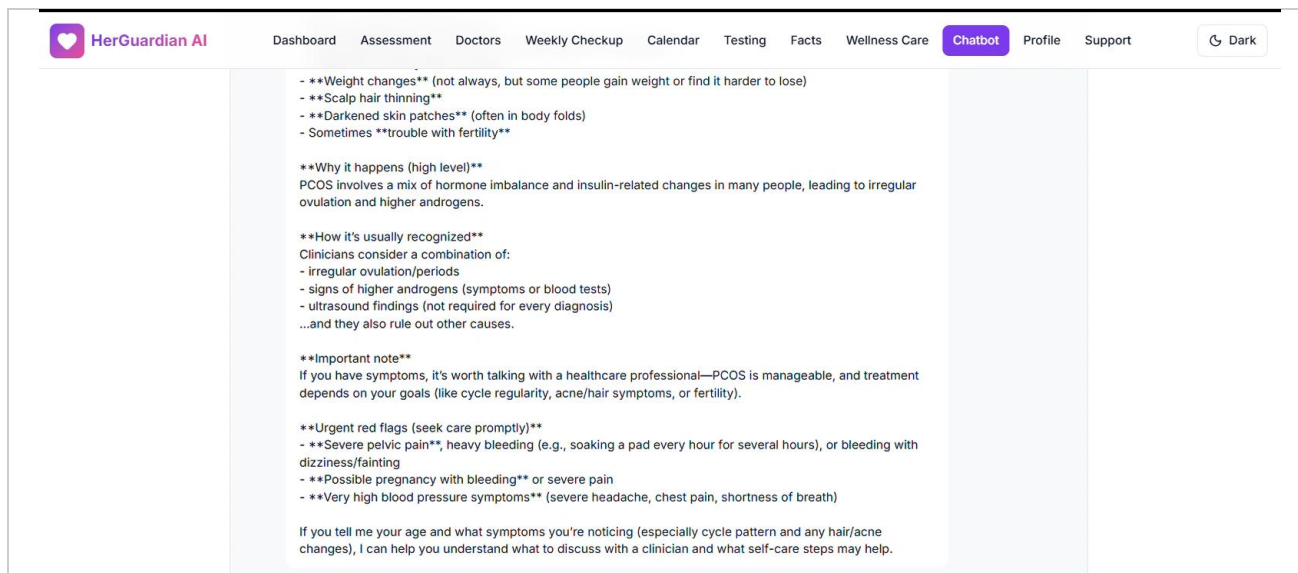


Figure 12 — Chatbot response (continued): safety note and PCOS-specific urgent red flags.

5.8 Support & Payments (Razorpay)

Sample Input: Payment option: UPI apps · Amount: ₹1 (test transaction)

Expected Output: A Razorpay order is created, checkout opens with the entered amount, and the user can pay using the selected method.

The Support page combines Razorpay-ready checkout options (UPI, UPI ID/QR, net banking, cards) with a help desk and first-checkup discount coupons:

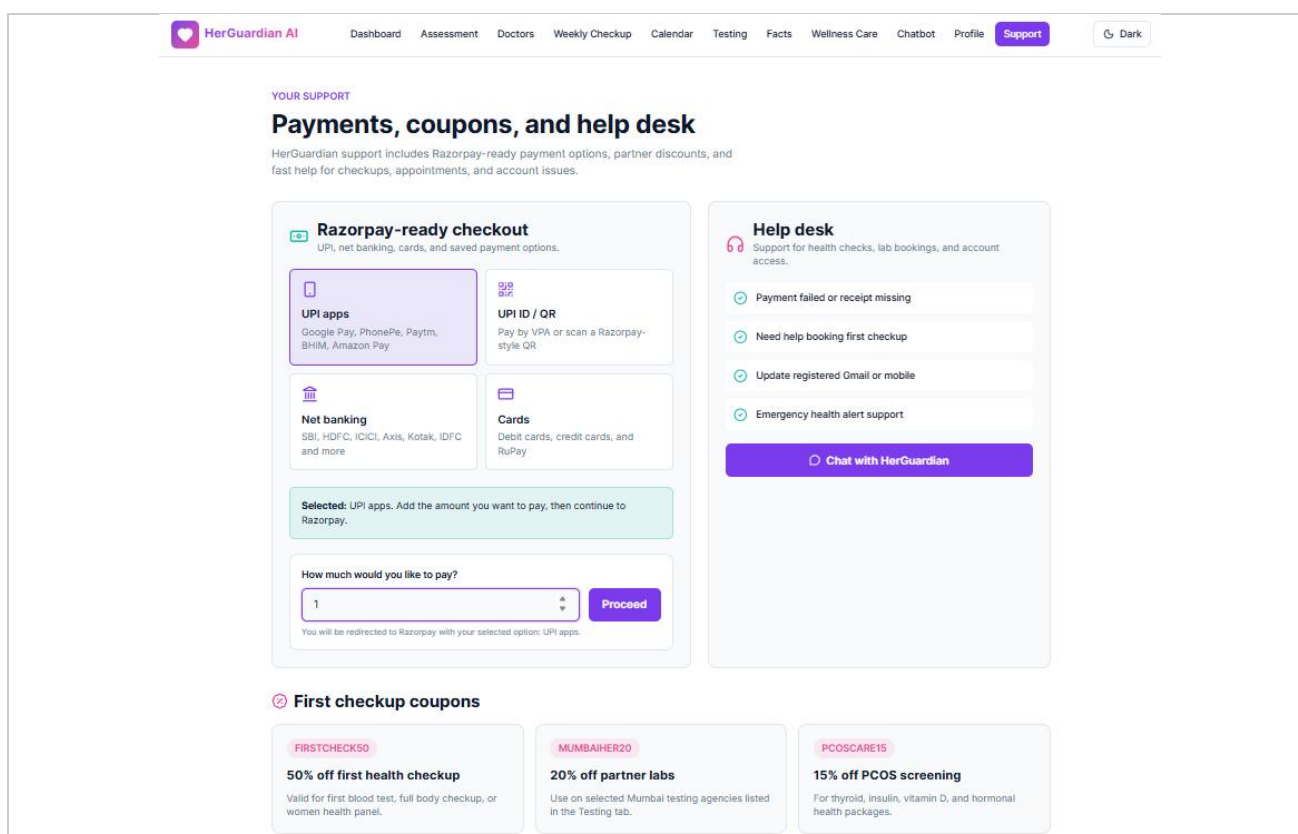


Figure 13 — Support page: payment method selector, help desk, and first-checkup coupons.

Selecting UPI apps and proceeding opens the live Razorpay checkout with the price summary and QR/UPI options:

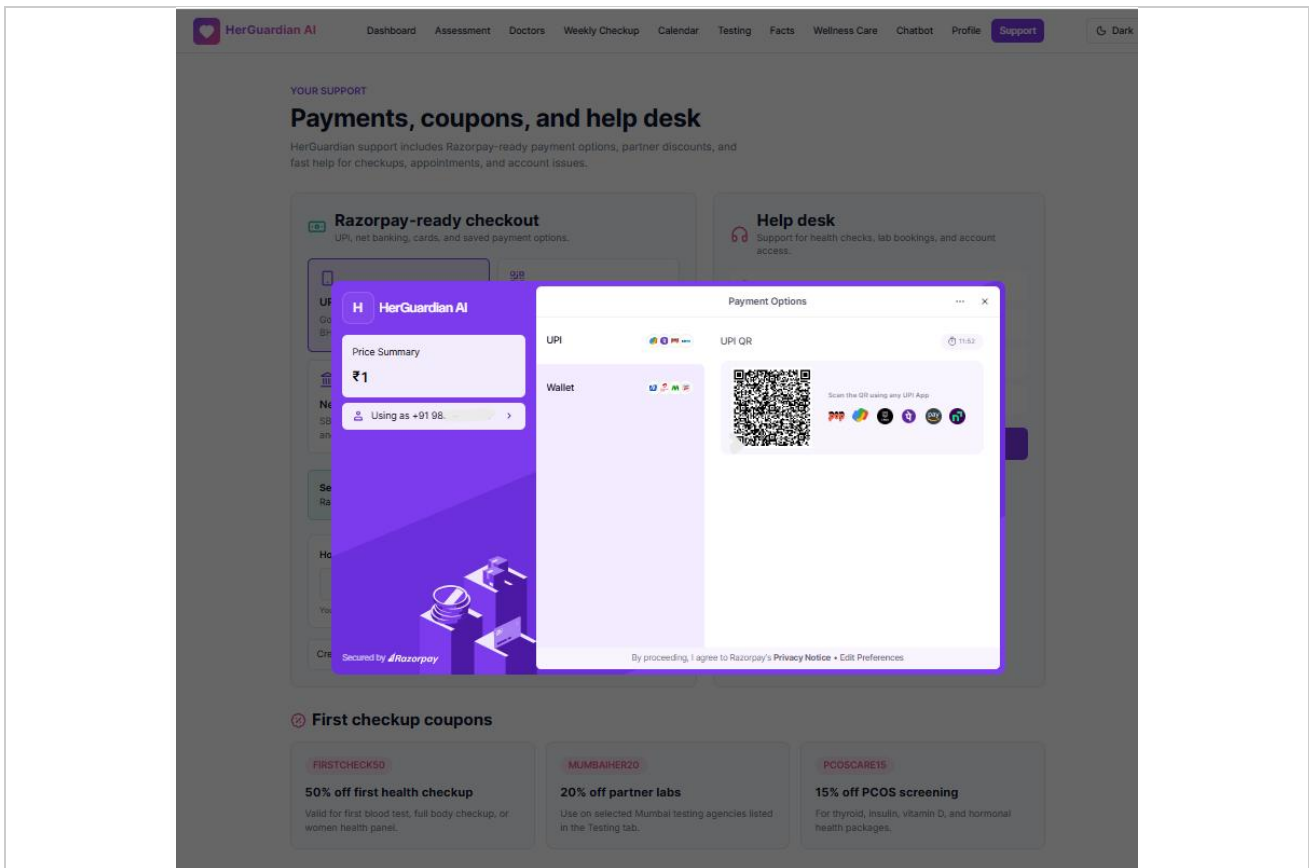


Figure 14 — Razorpay checkout modal showing UPI and UPI QR payment options for a ₹1 test order.

Completing the UPI payment returns a confirmed success state from Razorpay for the HerGuardian AI order:

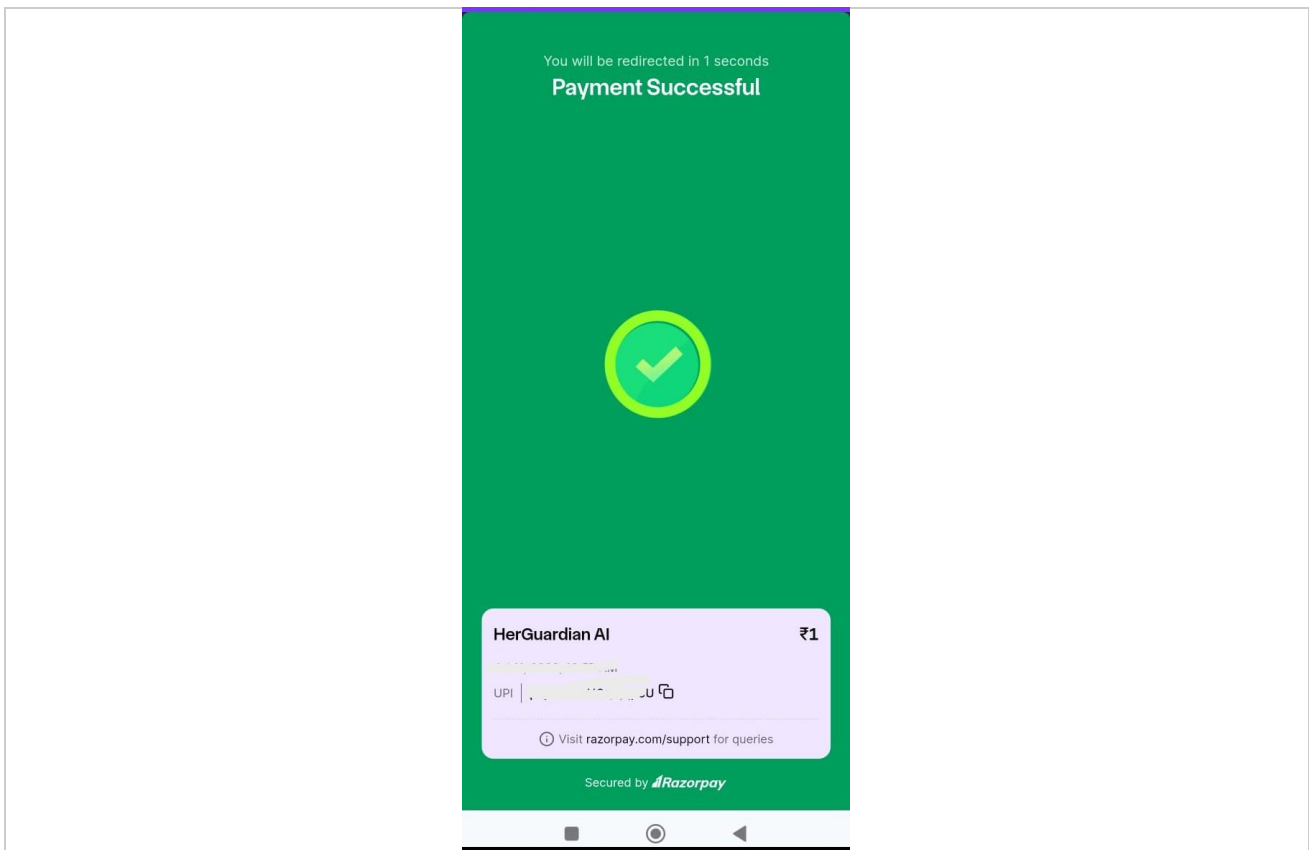


Figure 15 — Razorpay "Payment Successful" confirmation for the test transaction.

6. Medical Safety & Disclaimer

HerGuardian AI is designed to provide screening support, health education, preventive guidance, and timely awareness for women's wellbeing. It is explicitly not a diagnostic tool. The platform focuses on:

- Helping users understand possible health patterns and risk signals
- Encouraging early conversations with qualified healthcare professionals
- Supporting safer next steps through reports, reminders, and appointment booking
- Promoting preventive care, health literacy, and emergency awareness
- Keeping final medical decisions with certified doctors and healthcare providers

When serious symptoms or red flags are detected, the platform guides the user toward professional care, emergency support, or booking a doctor appointment directly through the app, as reflected in the chatbot and assessment screenshots above.

7. Conclusion

Across the assessment, doctor booking, testing, calendar, wellness-care, chatbot, and payment flows, the live deployment produced outputs consistent with the expected behaviour defined for each module. The system runs entirely on managed cloud infrastructure (Vercel, Neon PostgreSQL, external LLM providers, Razorpay, Twilio) with no local model dependency, keeping client-side resource usage light and inference latency dependent primarily on network and provider response time.

Further review — including clinical validation of risk-scoring logic and expanded automated testing — is recommended before any production healthcare use, consistent with the project's academic/hackathon scope stated in the README.